

CONSCIOUSNESS IN A BLOCK UNIVERSE: THE HYPOTHESIS OF TEMPORAL CORRELATION

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Abstract

This essay is not a scientific paper. It contains only the author's personal views on matters of human consciousness. Artificial intelligence (AI) was partially used to develop the ideas whilst writing this essay.

The distinction between the past, the present and the future is merely a stubbornly persistent illusion.

A.Einstein

Many people are familiar with the sensation of déjà vu (from the French déjà vu — “already seen”). For example, we are walking down the street and see familiar objects, sounds, conversations, and a thought flashes through our mind — “I’ve seen, heard and felt this before”. I remember one such episode from my childhood. Once, I was walking on the lawn next to a local shop. And suddenly I realized that I had seen all of this in a dream. I’m not just talking about the grass, the shop, the weather, but also the exact route, even though I always walked completely differently. And at that moment, it didn’t feel like “I remember all of this”, but rather “I’ve literally already been here, felt all of this in a dream”. And, like any curious young lad, an idea struck me. I wanted to change what I’d seen. I started walking in circles on the lawn, constantly changing my path, doing anything and everything just to behave unpredictably, so that what I’d seen couldn’t come true. But in the midst of my defiance, I felt that familiar sensation again. It was as if the next fragment of a memory of my dream had opened up to me, in which I was still walking in circles and changing direction haphazardly, trying to deceive myself.

As a child, I used to have dreams like that from time to time, which would then come true, but after that incident, I never tried to change my fate again. I realised that all I could do was prepare myself mentally for a particular event. Of course, almost all the dreams were forgotten upon waking, but in my mind, I intuitively sensed the possibility of certain events occurring. And, if they did happen, I would feel again — “I saw this too”. At the same time, a thought would even flit through my young child’s mind: “What if everything that happens is, to some extent, predetermined? If dreams come true, does that mean these events have already happened at some point in space-time, and my consciousness at the “now” point is merely catching up with these events, and in my dreams, I can somehow receive information about what I will see and hear in the future?” An illusion is created as if the soul has already experienced certain events, whilst the body and consciousness “now” are only yet to arrive at this point.

In my work, I wish to attempt a theoretical description of how such coincidences might occur.

I. THE BLOCK UNIVERSE MODEL

This paper proposes, drawing on the work of scholars in the fields of research into the characteristics of human consciousness and dreams, as well as assumptions regarding the possible structure of time, to put forward a hypothesis that reconciles my personal observations with the assumptions of researchers in these fields.

It is proposed to consider three positions that could form the basis of the hypothesis:

1. Strictly materialistic – everything is explained by memory, prediction and reconstruction.
2. Agnostic – we do not understand the nature of consciousness well enough to definitively rule out unusual effects of time perception.
3. Metaphysical – consciousness may be related to time in a way different from what we are accustomed to.

It is proposed to take as a basis the block universe model, where the past, present and future can be regarded as parts of a single space-time structure [1]. Minkowski space plays an important role here. The German mathematician argued that space and time form a single entity. This became the cornerstone of the special theory of relativity, allowing us to view the structure of the universe in a new light. It unites the three spatial dimensions and time into a single four-dimensional continuum [2]. In turn, Julian Barbour, a British theoretical physicist, argued that time may be an illusion, and reality merely a set of states [3]. The most incredible hypothesis was put forward by Kurt Gödel. He developed a solution to the equations of general relativity, in which the concept of objective time becomes strange and ambiguous. In his model, space-time turned out to be closed along a curve, which theoretically allowed for time travel, including a return to one's own past [4].

Summarising the work of scientists, one can put forward the hypothesis that the past, present and future already exist as a holistic structure, and consciousness sometimes touches upon its fragments. Based on this, we can identify the main points that should form part of its body:

1. The future already exists as a structure.

In a block universe, space-time is a single object; all events are “arranged” within a 4D structure, and the present is a cross-section of perception. Information from future cross-sections may partially influence the brain's current states.

2. Self-consistency.

Novikov's principle of self-consistency within the framework of the hypothesis can be interpreted as follows: “If information from the future exists, it cannot alter events that have already occurred”. That is, information does not change what has already happened, but is “embedded” within it.

3. Incomplete accessibility.

Information flowing through the complex structure of space-time must be fragmentary, noisy and incomplete.

4. Impossibility of arbitrary retrieval.

Otherwise, controlled telepathy and causality paradoxes would arise.

5. A possible class of mechanisms:

- *“closed information loops”*:

One way of theoretically substantiating the proposed hypothesis is to postulate the existence of closed information loops. The idea is that information does not arrive directly “from the future”; rather, it circulates within a closed structure in

time. The brain receives not the “future”, but a self-consistent fragment of information that is already part of the overall causal loop.

- *“weak quantum retro-correlation”*:

In quantum mechanics, there are interpretations where the state of a system depends not only on the past, but also on future boundary conditions. For example, ideas such as the two-vector formulation of quantum mechanics, or the transactional interpretation. In this logic, a quantum event is determined by the “encounter” of waves from the past and the future. Information is, as it were, determined from both sides of time. In this case, the brain does not “see the future” directly, but receives statistically corrected patterns in which the future is already “factored in”.

- *“informational resonance synchronisation”*:

The brain can be viewed as a complex dynamic system that constantly predicts the state of the world. This class assumes that the Universe is not only matter, but also a global information structure, where future states already have a mathematical description, and the brain can sometimes resonate with these states.

It can be described by the following mechanism:

1. The brain constructs a multitude of models of the future.
2. Some of these coincide with the actual future state.
3. When the match is sufficiently close, a strong sense of déjà vu arises.
4. At the same time, memory may “fill in” the details retrospectively.

This resembles not the transmission of information from the future, but the selection of the correct solution from an already existing space of possible trajectories.

Even if we accept the existence of a block universe model, this does not imply the possibility of accessing future states. However, if consciousness is part of this structure, the question arises as to the possibility of correlation between different sections of a single timeline.

II. CONSCIOUSNESS AS A DYNAMIC SYSTEM

It is worth noting here the current scientific view on the subject of vivid emotional dreams in children. Adolescence is a period during which the prefrontal cortex (which is responsible for control and reality testing) undergoes active restructuring. Memory filtering intensifies, sleep patterns and REM phases change, and the “sense of time” and causality stabilise [5]. Consequently, adolescents experience more vivid dreams and a greater number of hyper-detailed images.

A child’s brain is more plastic, more immersed in figurative thinking, processes memory and time differently, and has a less stable boundary between imagination, sleep, memory and expectation. As we grow older, memory becomes more structured, the perception of time becomes more stable, and the brain retains dream fragments less effectively.

With age, there is a decline in “dream-like richness”. People experience fewer vivid REM dreams; these are less easily remembered, and there is greater fragmentation. The mature brain checks “did this happen or did I make it up? ”, attempting to suppress false coincidences.

The sensation that the soul seems to leave the body and head towards the very same body at another point in space-time can be linked to an age-old philosophical problem: “What is the

present? Why do we move through time, and must consciousness be strictly localised at a single point in time? ” We always experience the “now” as a single point, but the brain is constantly comparing “expectation” and “reality”. When a coincidence occurs, there is a sudden sense that the scenario has come to a close. This can be subjectively interpreted as a transition between already existing states. But in physical terms, it remains a change in the states of the neural system, rather than a movement of the observer through time.

The Swedish psychologist and psychiatrist Carl Jung suggested that the psyche may be connected to events not only causally; there are meaningful coincidences (synchronicity). Synchronicity is a concept that describes coincidences between external and internal events, unconnected by a cause-and-effect relationship, but united by a resonance of meaning. These are not mere coincidences, but moments when internal psychological states (thoughts, feelings, dreams, premonitions) are unexpectedly reflected in the external world [6]. John Dunn argued that people sometimes see future events in their dreams, that consciousness can perceive time non-linearly, and that dreams blend the past and the future [7]. Dunn’s theory holds that, in reality, the past, present and future, in a certain sense, occur “together”. Human consciousness, however, perceives this simultaneity in a linear form. In the state of sleep, the perception of time loses the linearity inherent in it during wakefulness. As a result, a person acquires the ability to have precognitive dreams, in which consciousness freely crosses the boundaries of the past, present and future.

From a physical point of view, brain activity is realised through electrical and chemical processes. These manifest as neuronal activity, that is to say, they constitute a dynamic system (fields, impulses, connections). In turn, this system is sensitive to states (sleep, stress, attention, memory). In other words, the brain is not a “storage facility”, but a process within a physical system. However, for it to be possible to obtain information about the probable future, a radical assumption is required: the brain is not a closed system in terms of information, that is, it can receive weak corrections from the outside or be part of a larger temporal structure.

Within the framework of a block universe, the present is simply a local cross-section of perception. The idea behind the hypothesis is that the brain sometimes “reads” neighbouring layers of the spacetime structure. Not via a signal, but via a correlation of states. This is not a transmission of information, but a coincidence of the brain’s structure with an already existing fragment of the future state of the universe. In other words, there is a single structure of time, and a person, as an observer, “glides” across its various cross-sections.

III. INTRODUCTION TO THE CORRELATION HYPOTHESIS

The hypothesis under consideration is grounded in contemporary scientific ideas, such as: consciousness as a dynamic system rather than an object; the brain as a predictor of the future; time as a complex structure.

From a scientific point of view, the main idea can be formulated as follows. Consciousness can access an already existing configuration of future states and experiences it as a memory. The brain generates predictive models of the future, which sometimes coincide with reality, and the area of coincidence is perceived as a memory. Within the framework of this hypothesis, future states are viewed not as a single fixed line, but as a structure of potential trajectories of varying degrees of stability.

We can now formulate a hypothesis regarding global temporal correlation: “If we assume that consciousness is a physical system of fields, then the brain can sometimes enter a state that

coincides with the trajectory of a future state of the universe, and this coincidence is subjectively experienced as a memory of the future”.

My experience, in the context of the block universe model, implies that consciousness sometimes gains access not to a “prediction”, but to an already existing segment of its own worldline. Thus, a person does not “see the future”, but traverses it as a region of space-time. If one follows the model literally, then during sleep a person is in a state that correlates with the future state of the brain. In other words, the brain at point A “resonates” with itself at point B (in the future). This is not a transfer of information, but rather the coincidence of two cross-sections of the same four-dimensional structure.

In that case, the “law of immutability” becomes natural, because the person is already on a trajectory where any attempt at change is already part of it. This corresponds to the idea of self-consistency. That is, any “resistance to the scenario” is already built into the scenario. In that case, “dreams” are not a preview of the future, but local projections of the entire branch. According to the hypothesis, dream fragments may not be random generations, but rather “encounters” with one’s own global timeline. Consequently, they may represent a brief coherence between the current state of the brain and the future state of the same system.

IV. DESTINY AND CHOICE

If we assume the existence of a world-line within the structure of space-time, it may turn out that a person’s entire life is strictly predetermined. This approach gives rise to a number of philosophical implications. For example, the very concept of free will is altered. A person does not “choose the future”, but realises an already existing trajectory geometry. They cease to be an observer in time, and become a local process within a temporal object, which becomes self-consistent with its other segments.

So, if “fate” is a world line, let “human life” be a single branch within this object. Then “fate” is simply a fixed world line in space-time.

In that case, what is “choice”?

1. Option A (hard determinism). Choice is an illusion. The state of the brain at time t completely determines the state at $t+1$. No alternative “really” exists. Consequently, formally speaking, a person does not choose, but realises an already existing path.
2. Option B (compatible with quantum physics). Choice is a point of branching of probabilities, but only in perception. At the level of microphysics there is uncertainty, but the whole story is still “put together in its entirety”. Then “choice” is a local sensation of transition along an already existing branch.

Within this hypothesis, dreams become a special phenomenon: a local synchronisation of the current state of the brain with the future state of the same worldline. In other words, the brain does not “guess”; it sometimes temporarily enters a mode of alignment with another point on its own branch.

If we accept this assumption, a new classification of states of consciousness emerges:

1. Local state (ordinary wakefulness).
2. Predictive state (thinking).
3. Resonant state (synchronisation of consciousness across different time scales).

To prevent the model of the hypothesis from collapsing, an important constraint is introduced: temporal information asymmetry. Even if the future structure exists, access to it is

not symmetrical, not arbitrary and not controllable. Otherwise, information loops would arise, causality would be violated, and the possibility of changing direction would emerge. Therefore, resonance must be rare, uncontrollable and not amplifiable at will.

So, what is a “choice error”? Within the hypothesis, an error is not a “deviation from destiny”, but a transition between already existing permissible states of the trajectory. In other words, a person does not step outside the bounds of destiny, but moves within its internal structure.

If one accepts this system to its logical conclusion, a complex situation arises: morality and responsibility remain, but their meaning changes. Because actions are still causally linked, consequences still materialise, and the subjective “I decided” remains part of the process. In other words, even if the choice is “predetermined”, the experience of choice remains a functionally necessary element of the path.

Within the framework of this hypothesis, it is proposed to consider several possible ways in which the trajectory of the world-line might be represented within the structure of space-time:

1. A rigid, fixed path (fate is absolute).

This is the “purest” form of the block universe. How it works: a person’s entire life already exists in its entirety. Every choice is already built into the structure. There were never any alternatives. Consequently, they “could not have acted differently”. The sense of choice is part of the journey. Even doubts are already predetermined. But the price of this hypothesis in this concept is that freedom of choice becomes an illusion.

2. A branching structure (the soft version).

The model of the hypothesis moves closer to quantum interpretations. How it looks: there are many potential paths. Consciousness moves along one of them. Choice is the moment a branch is fixed. In this case, dreams may be a glimpse of the most probable branch, but not a guarantee. This explains why some dreams come true, and some did not.

3. Hybrid version.

The global structure is fixed, but within it there are local areas of uncertainty. That is, the key nodes of destiny are unchangeable, whilst the minor details are variable. My experience can then be explained as follows: I saw the “anchor node” of the line, not every detail. And when I tried to “change” events, I still returned to the same node in the structure.

The first version is highly internally consistent; however, it completely eliminates freedom of choice. Further discussion will focus on the development of the second and third models. Within these frameworks, if dreams sometimes do not correspond to reality, it means that the person saw one of the possible outcomes, but subsequently made a choice that simply did not lead to it. It follows, then, that there are not “infinite chaotic worlds”, but a set of potential pathways. Some of these are more stable and probable. This means that, in a dream, consciousness can “resonate” not only with the line that was ultimately realised, but also with one of the possible ones. In that case, a dream that comes true is contact with the realised branch, and a dream that does not come true is contact with a potential but unchosen trajectory. This version preserves the sense of destiny, but does not completely negate the concept of choice. That is, “choice” can be defined as: the moment when the system settles on one of the possible paths.

In this version, the law of immutability can also be explained. “What has been seen cannot be changed” not because change is impossible in general, but because if a sufficiently strong resonance has occurred, it means the line was already very stable and the probability of

“deviating” from it was minimal. Therefore, any attempts to “change” things will still lead back to the same node of events. Consequently, a very interesting idea emerges: consciousness does not create the future, but “passes through” the space of probabilistic structures. And dreams, in this case, are not prophecies, but a brief touch with neighbouring possible paths.

Here, the model effectively assumes that the future exists in part as a space of potentialities, rather than as a single rigid sequence. This is no longer pure eternalism nor a purely quantum multidimensional interpretation, but a set of possibilities within which consciousness follows a specific path.

The most interesting consequence is that time ceases to be a linear flow; it becomes a “space of variants”. And consciousness becomes a mechanism for navigating within this space.

V. STRUCTURE OF THE HYPOTHESIS OF TEMPORAL CORRELATION

Summarising the above and drawing on personal experience and the work of scientists, whilst taking into account the assumptions and limitations introduced, the hypothesis of temporal correlation can be formulated as follows:

1. The structure of the Universe.

The Universe is not a linear flow of events, but a multidimensional structure of potential trajectories. Time does not “flow”. There exists only a space of states and the connections between them. The past, present and future – are not fleeting moments, but elements of a unified structure.

2. Consciousness.

Consciousness is not an object within time. It is a process of sequentially passing through the space of possible states. A person experiences this movement as the “present”, the passage of time and personal choice.

3. Choice.

Choice does not create the Universe from scratch nor does it violate causality. It fixes one of the available directions within the space of possibilities. Therefore, fate exists as a set of permissible paths, but the specific route is determined by a sequence of choices.

4. Dreams and a glimpse of the future.

In a dream, the mind’s rigid fixation on the current state of reality is weakened. At such moments, the mind can resonate with neighbouring or future potential states of its own trajectory. This is experienced as prophetic dreams, a sense of familiarity, déjà vu, a feeling of certainty.

5. Why do some dreams not come true?

Not all the states we see are definitively realised. Some of them are merely probable branches that were not chosen. Therefore, dreams that come true are a touch upon a realised branch. Those that do not come true are a touch upon a possible, but unchosen, path.

6. Why are some events impossible to change?

There are nodal states – points of high trajectory stability. As one approaches them, the space of possibilities narrows, and any attempts to deviate still lead to the same outcome.

7. What is déjà vu?

Déjà vu is not a memory error, but a moment of re-encountering a state with which consciousness has already resonated previously. A person does not “remember the past”. They recognise the structure of a path they have already touched upon.

Thus, the main principle of the hypothesis can be formulated: Time is not a river. The future is not a void. Consciousness is not a passive observer. The universe is a field of possible trajectories, and human life is a process of navigation within it. A person does not move from the past into the future. Perhaps they gradually uncover the already existing structure of their own reality – and sometimes, in dreams or in rare states of consciousness, manage to glimpse fragments of the path before they reach them.

On the use of artificial intelligence in writing this work.

AI was used as a tool for literature searches, discussing hypotheses and obtaining recommendations on text structure. The main concept, argumentation and final editing of the work are the author's own.

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